

Professional Summary

Cell biologist with 10 years experience managing multiple research projects in the following areas:

- 6 years experience in membrane trafficking and heterotrimeric G-protein signaling
- 4 years experience in cell shape and motility, G-protein signaling, and mitochondria biochemistry

Education

University of California, San Francisco, Ph.D. Biomedical Sciences (focus: Cell Biology and Pharmacology)	2017-2023
Yale University, Masters Program in Biological Sciences	2015-2017
University of Maryland, B.A. Physiology	2010-2014

Research and Professional Experience

Department of Medicine, Yale University

Mentor: Ansel Adams, Ph.D.

Postdoctoral Research Fellow

2023-Present

- **Project:** Investigate the role of plasma membrane transporters and EM proteins in multiple cell functions including cell motility, cell shape determination, invasion, proliferation, and survival. Opened a new field of research related to the bifunctionality of membrane anchoring proteins.
- **Responsibilities:** Responsible for design, implementation, and management of multiple research projects; Concurrently advised three junior postdoctoral fellows; Supervise and direct the research projects of graduate and medical students; Review grant proposals and manuscript submissions; Train and supervise technical and research staff; Assist in the hiring of technical staff; Teaching Assistant, Cell Biology. Led ad hoc discussion section; Set up time-lapse video system.
- **New Methodologies:** time lapse video and live cell fluorescence imaging, DNA Microarray analysis, adhesion and invasion assays, kinase assays, FACS and TUNEL analysis.

Program in Biomedical Sciences, University of California, San Francisco

Dissertation chair: Georgia O'Keefe, M.D.

Doctoral Research

2018-2023

- **Project:** Investigated the function of alpha-example and beta-example coupled receptors in mitochondria membranes in the rat kidney. Developed a biochemical purification strategy that led to the discovery that X and Y subunits of alpha-example are differentially expressed on mitochondria membranes.
- **Responsibilities:** Developed a strategy for bulk separation of plasma and mitochondria membranes based on subcellular fractionation using sucrose gradients; Provided research advice and supervision to junior graduate and undergraduate students; Developed a lectin-based system for the removal of contaminating plasma membranes from mitochondria membranes; Teaching Assistant, Pharmacology and Toxicology. Assisted in course development, led discussion sections and lectured on Basic Principles of Toxicology.
- **New Methodologies:** animal surgery, tissue sectioning, subcellular fractionation, antibody production, yeast two-hybrid analysis, bacterial and yeast cell culture, protein purification, biochemical and molecular techniques

Center for Biomedical Research on Aging, Yale University

Advisor: Charles Monet, M.D.

Research Assistant

2016-2017

- **Project:** Compared the S-Sample Receptor heterotrimeric G-protein signaling pathways in primary osteoblasts to that of osteoblast cell lines and found differential sensitivities to PTH.
- **Responsibilities:** Obtained bones from fetal mice; Used enzyme digestion to generate osteoblasts for primary cell culture; Graduate Assistant, *Human Anatomy*: Organized study material, lectured, and led discussion sections.
- **New Methodologies:** microdissection, immunocytochemistry, adenylyl cyclase assays, cell culture.

Immunology & Environmental Medicine Clinic, College Park, Maryland

Employer: Smokey Bear, M.D.

Clinical Technician

2014-2015

- **Responsibilities:** Performed and analyzed allergen sensitivity exams; Taught patients self-injection techniques; Listened to patients discuss allergen sensitivities and worked with them to develop lifestyles to minimize their exposures; Prepared antigen solutions by serial dilution.

Publications

Invited

Muir, J., Earp, W., and London, J., 2025. The role of plasma membrane transporters in cell motility. In preparation for Current Opinions in Science.

Earp, W., Muir, J., and London, J., 2025. The role of EM proteins in determining cell shape. In: Annual Review of Pharmacology and Toxicology (L. Baranov, A, eds.), 23:134-147, Annual Review, Palo Alto, CA.

Reviewed

Muir, J., Earp, W., and London, J., 2024. Plasma membrane transporters in cell invasion. Molecular Cell 13:112-114.

Muir, J., and London, J., 2023. EM protein function. (Baranov, A, eds.), 22:124-129, Academic Press, San Diego, CA.

In Preparation

Muir, J. and London, J. Bifunctionality of membrane anchoring proteins. A description of this current project together with a Quicktime movie of mitochondria migrating into a wound can be viewed at: <http://www.sciencevideo.com/xhf17si3>

Earp, W., Muir, J., and London, J., New methodologies for investigating role of EM proteins in cell plasma membranes.

Meeting Abstracts

Muir, J. and Earp, W., The role of plasma membrane transporters in cell motility (ASCB Meeting in Washington, D.C., December 2024). Mol. Biol. Cell (S12): 2025.

Muir, J. and Earp, W., The role of EM proteins in determining cell shape (French-American Colloquium on the Mitochondria and Human Disease, Universite de la Mediterranee, Marseille, France, 2024).

Muir, J. and Earp, W., The role of plasma membrane transporters in cell proliferation and survival (Keystone Meeting: The Dynamics of the Cytoskeleton, Keystone, CO, 2024).

Wyneberg, T. and Earp, W., Plasma membrane transporters and EM proteins in mitochondria (ASCB Meeting in San Francisco, CA, December 2023). Mol. Biol. Cell (S7): 2023.

Mentoring Experience

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| • Present, Molecular Cell Biology undergraduate | AWIS (California Chapter) mentoring program |
| • 2016, Yale Dental Student | Role of proteins in cell adhesion and migration |
| • 2016, Yale Biomedical Sciences Graduate Student | Role of apoptosis in the proliferative action of adhesion |
| • 2016, Yale Medical Student | Association of actin-binding proteins with the plasma membrane |

Invited Conference Oral Presentations

- Mini symposium on Cell Adhesion, American Society for Cell Biology Annual Meeting, Washington, DC, 2025
- French-American Colloquium on the Mitochondria and Human Disease, Marseille, France, 2024
- Bay Area Research in the Cytoskeleton, 2024
- Mini symposium on Small GTPases and Control of the Cytoskeleton, American Society for Cell Biology Annual Meeting, San Francisco, CA 2023

Committees

- Judge, Siemens Westinghouse National Science & Technology Competition, 2024-present
- Future of the Plaza Coalition: A neighborhood development committee, 2020-2023
- Admissions Committee, Student Representative, Graduate Program in Biomedical Sciences, 2019-2023

Honors and Awards

- NIH Travel Award, Mitochondria and Human Disease, French-America Colloquium, France, 2024
- NIH Postdoctoral Training Grant, 2023-present
- Syntex Fellowship in Molecular Pharmacology, 2022-2023
- NIH Predoctoral Traineeship in Cellular and Molecular Pharmacology, 2021-2022
- Graduate Equity Fellowship, 2019-2023
- Faculty/Staff Nominee, Outstanding Senior Award, 2017